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...
Dataset loaded successfully
Dataset path: /content/drive/MyDrive/dataanalytics/Womens Clothing E-Commerce Reviews.csv
Dataset shape: (23486, 11)

First five records:

```

|   | Unnamed: 0 | Clothing ID | Age | Title                   | \ |
|---|------------|-------------|-----|-------------------------|---|
| 0 | 0          | 767         | 33  | NaN                     |   |
| 1 | 1          | 1080        | 34  | NaN                     |   |
| 2 | 2          | 1077        | 60  | Some major design flaws |   |
| 3 | 3          | 1049        | 50  | My favorite buy!        |   |
| 4 | 4          | 847         | 47  | Flattering shirt        |   |

```


```

|   | Review Text                                       | Rating | Recommended | IND | \ |
|---|---------------------------------------------------|--------|-------------|-----|---|
| 0 | Absolutely wonderful - silky and sexy and comf... | 4      |             | 1   |   |
| 1 | Love this dress! it's sooo pretty. i happene...   | 5      |             | 1   |   |
| 2 | I had such high hopes for this dress and reall... | 3      |             | 0   |   |
| 3 | I love, love, love this jumpsuit. it's fun, fl... | 5      |             | 1   |   |
| 4 | This shirt is very flattering to all due to th... | 5      |             | 1   |   |

```


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|   | Positive Feedback Count | Division Name  | Department Name | Class Name |
|---|-------------------------|----------------|-----------------|------------|
| 0 | 0                       | Initmates      | Intimate        | Intimates  |
| 1 | 4                       | General        | Dresses         | Dresses    |
| 2 | 0                       | General        | Dresses         | Dresses    |
| 3 | 0                       | General Petite | Bottoms         | Pants      |
| 4 | 6                       | General        | Tops            | Blouses    |

Missing values before cleaning:

|                         |      |
|-------------------------|------|
| Unnamed: 0              | 0    |
| Clothing ID             | 0    |
| Age                     | 0    |
| Title                   | 3810 |
| Review Text             | 845  |
| Rating                  | 0    |
| Recommended IND         | 0    |
| Positive Feedback Count | 0    |
| Division Name           | 14   |
| Department Name         | 14   |
| Class Name              | 14   |
| dtype: int64            |      |

Duplicate rows before cleaning: 0

Missing values after cleaning:

|                         |   |
|-------------------------|---|
| Clothing ID             | 0 |
| Age                     | 0 |
| Title                   | 0 |
| Review Text             | 0 |
| Rating                  | 0 |
| Recommended IND         | 0 |
| Positive Feedback Count | 0 |
| Division Name           | 0 |
| Department Name         | 0 |
| Class Name              | 0 |
| Full Review             | 0 |
| Cleaned Review          | 0 |
| Review Length           | 0 |
| Word Count              | 0 |

```

Dataset information after cleaning:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 23486 entries, 0 to 23485
Data columns (total 16 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Clothing ID                          23486 non-null  int64
1   Age                                  23486 non-null  int64
2   Title                                23486 non-null  object
3   Review Text                          23486 non-null  object
4   Rating                              23486 non-null  int64
5   Recommended IND                      23486 non-null  int64
6   Positive Feedback Count              23486 non-null  int64
7   Division Name                        23486 non-null  object
8   Department Name                      23486 non-null  object
9   Class Name                           23486 non-null  object
10  Full Review                          23486 non-null  object
11  Cleaned Review                       23486 non-null  object
12  Review Length                        23486 non-null  int64
13  Word Count                           23486 non-null  int64
14  Character Count                      23486 non-null  int64
15  Average Word Length                  23486 non-null  float64
dtypes: float64(1), int64(8), object(7)
memory usage: 2.9+ MB
None

```

```

▶ X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y,
)

tfidf_vectorizer = TfidfVectorizer(
    max_features=7000,
    min_df=3,
    max_df=0.85,
    ngram_range=(1, 2),
)

print("\nTraining records:", X_train.shape[0])
print("Testing records:", X_test.shape[0])

...
Training records: 18788
Testing records: 4698

```

### Top Chi-Square text features:

|      | Feature       | Chi Square Score |
|------|---------------|------------------|
| 233  | disappointed  | 138.062061       |
| 1111 | wanted love   | 135.266834       |
| 1085 | unflattering  | 116.475370       |
| 771  | poor          | 107.073673       |
| 483  | huge          | 93.148517        |
| 130  | cheap         | 90.953492        |
| 411  | going back    | 85.659308        |
| 1108 | wanted        | 84.337936        |
| 837  | returned      | 81.876849        |
| 578  | looked        | 75.772190        |
| 838  | returning     | 75.510425        |
| 774  | poor quality  | 63.523552        |
| 238  | disappointing | 63.437257        |
| 1088 | unfortunately | 61.364095        |
| 824  | really wanted | 59.094521        |
| 581  | looked like   | 56.798705        |
| 50   | awful         | 54.964172        |
| 60   | bad           | 51.200595        |
| 1110 | wanted like   | 48.055310        |
| 1017 | terrible      | 47.643526        |

### Most positively reviewed departments:

```
Department Name
Unknown      100.000000
Intimate      80.922190
Jackets       80.620155
Bottoms       80.494867
Tops          76.709973
Name: Positive, dtype: float64
```

### Most negatively reviewed departments:

```
Department Name
Trend         18.487395
Dresses       10.903624
Tops          10.641956
Jackets       10.562016
Intimate       8.818444
Name: Negative, dtype: float64
```

Conclusion after business analysis: The review dataset shows strong overall customer recommendation behavior, but performance varies by department, class, age group, review length,